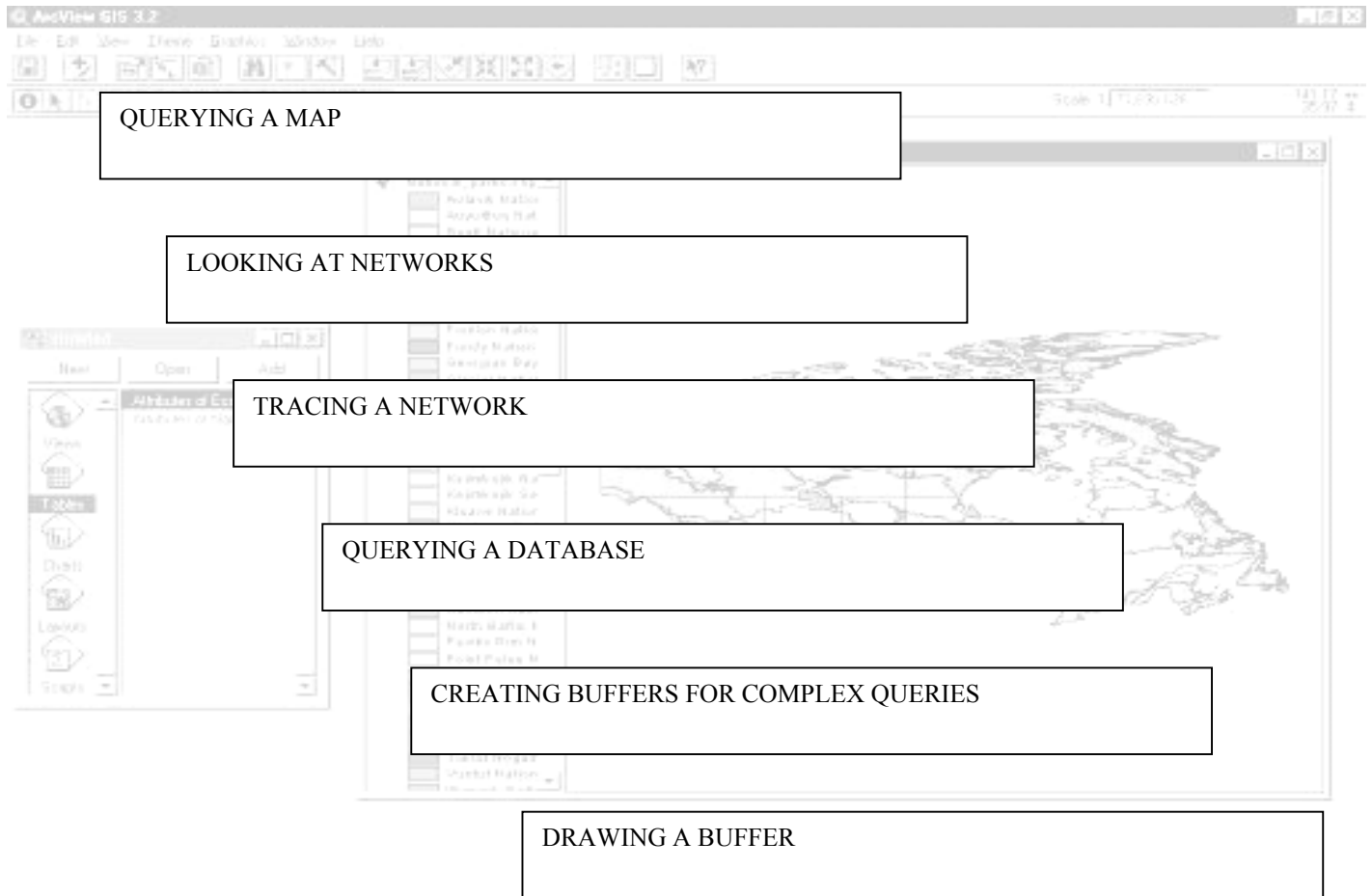


Transportation Trends in Canada



GIS Information

Here's a list and explanation of important features in ArcView:

Project : In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

View  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

Tables  : Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

Layout  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View – Tools

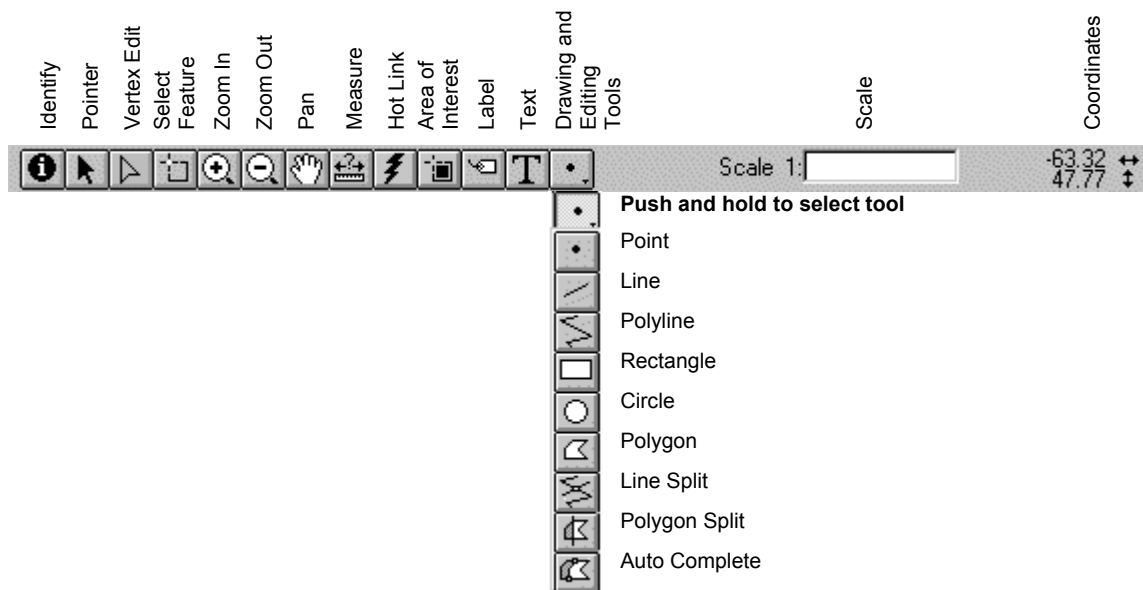


Table - Buttons



Table - Tools



Chart - Buttons

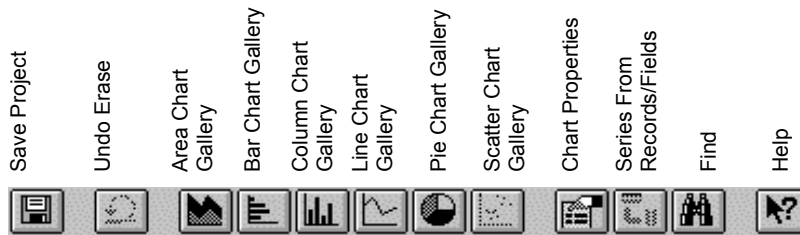
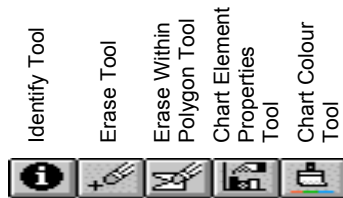
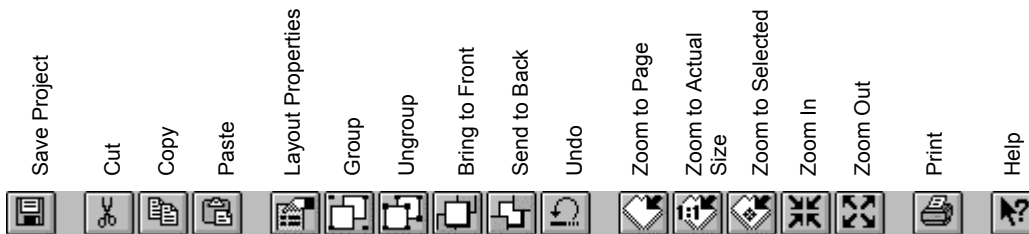


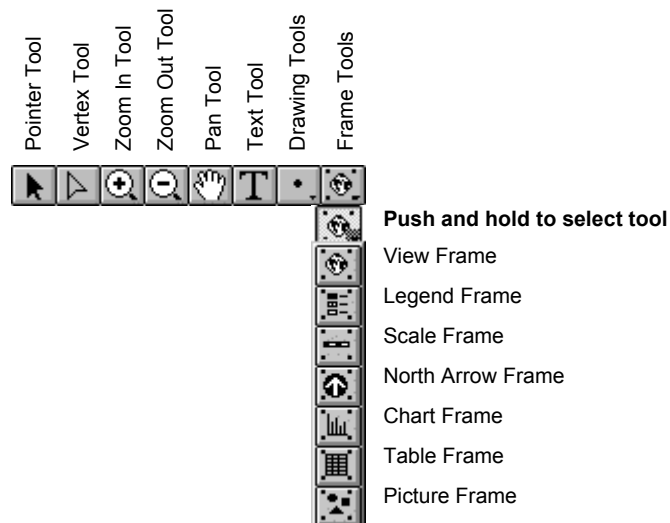
Chart - Tools



Layout - Buttons



Layout - Tools



For more Information: <http://www.esricanada.com>

STARTING THE PROJECT

In this lesson we will examine the distribution patterns of transportation systems. Software functions of query, statistics, buffer creation and length measurement will be used.

STARTING ArcView

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*.



2. Click **O.K.**

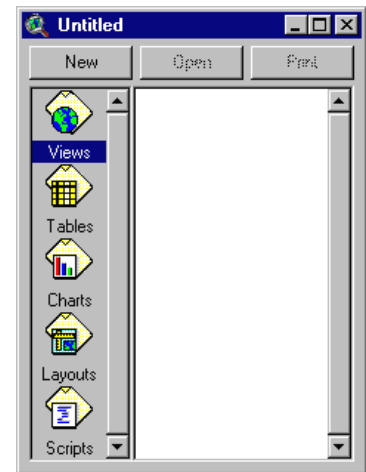
When the program starts, it automatically opens a project window titled "**Untitled.**"

If you like, you can maximize the **ArcView** window to fill the screen.

! In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.

3. Now you are ready to add documents to your new project.

4. With the "**Views**" icon selected, click on **New** and maximize the **View1** window.

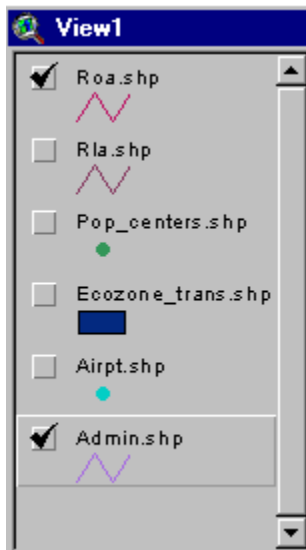


5. Click the **Add Theme**  button and navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

Admin.shp
 Ecozone_trans.shp
 Roa.shp
 Rla.shp
 Airpt.shp
 Pop_centers.shp

6. Click **O.K.**

This will add these six themes to the table of contents of the new view. They are now available for display and editing.



7. Click and drag to move the themes up or down in the table of content to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line and point themes should be on top of polygon files to be visible.

8. Click on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be changed by clicking on other themes.

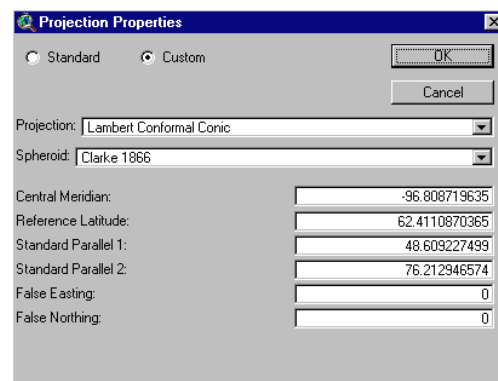
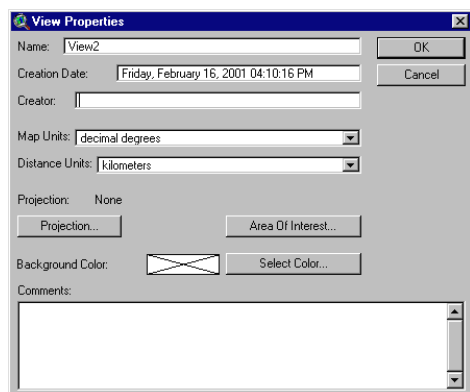
9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project**. Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (the suffix **.apr** means that you are saving your work as a project that can be opened up in the future).

ArcView - Changing the Map Projection

! A **Map Projection** defines the relationship between the map coordinates and the geographic coordinates, (latitude and longitude). Our current display shows what a map of Canada looks like with only a geographic projection.

Now we will give our map “projection coordinates” to make it look more like the maps which are familiar to us.

1. From the **View** menu, choose **Properties**.
2. Select **Properties**.
3. Select **Map Units** list. **Change the map units to decimal degrees**.
4. Select **Distance Units** list. **Change the distance units to kilometers**
5. Click the **Projection** button.
6. Select **Custom**.
7. Select **Lambert Conformal Conic** from the projection list.



! Take a look at the parameters for this projection. Parameters are the descriptions of how the projection was made.

Note that a spheroid was used. (The earth is a slightly flattened sphere toward the poles). **Conformal** implies that all angles are indicated correctly. All angles measured on the earth's surface are given the same values on the map.

Conical means in the shape of a cone. In this projection, the relative positions on a globe are transferred to a plane in the shape of a cone around the globe. See how the longitudinal lines are closer together at the top of the image, that is, they are not parallel.

The **central meridian**, **false easting** and **false northing** are used to define the X (easting) and Y (northing) coordinates of the map. The **reference latitude and standard parallel 1 and 2** are used to determine the origin of the projection (i.e. the point from which you are viewing the globe).

7. Click **O.K.**

The bottom of the **View Properties** window should look like this:



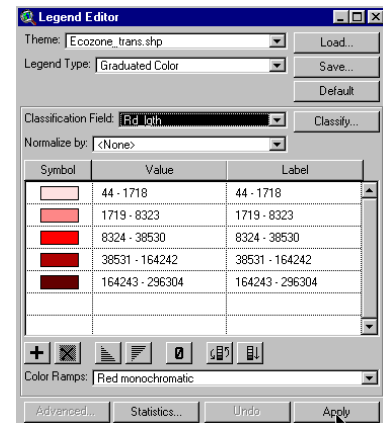
9. Click **O.K.**

! Note the difference in the map presentation.

? How is it different from the non-referenced map?

ArcView - QUERYING THE MAP

1. Double click **ecozone_trans** theme.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.
5. Select **Rd_lgth**. (road length).
6. Click **Apply**.
7. Close **Legend Editor** window.

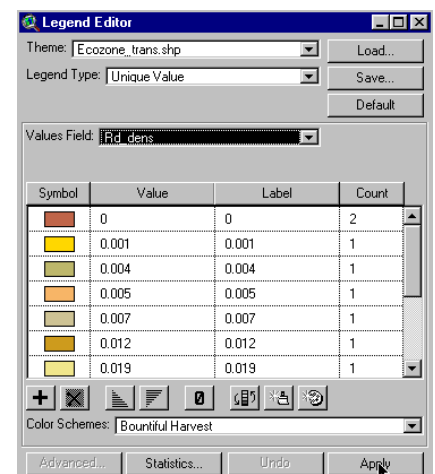


8. Select the **Identify Tool**.  Move the cursor over each ecozone and click to get information on that ecozone.

? Which ecozone has the greatest length of roads?

? Which ecozone has the second greatest length of roads?

? Is this what you would expect? Explain.



9. Double click **ecozone_trans** theme.
10. Open **Classification Field** list.
11. Select **Rd_dens** (road density).
12. Click **Apply**.
13. Close **Legend Editor** window.
14. Click the **Identify Tool**.  Move the cursor over each ecozone and click to get information on that ecozone.

- ? Find the ecozone with the highest road density?**
- ? Which ecozones have the second and third highest densities?**
- List two pros and cons related to high road density.

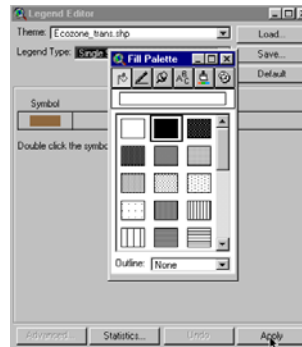
ArcView - LOOKING AT NETWORKS

One definition of a network is an arrangement of intersecting lines and nodes. Transportation networks can be made up of roads and rail lines that intersect at many locations which are often points of commerce such as cities and ports.

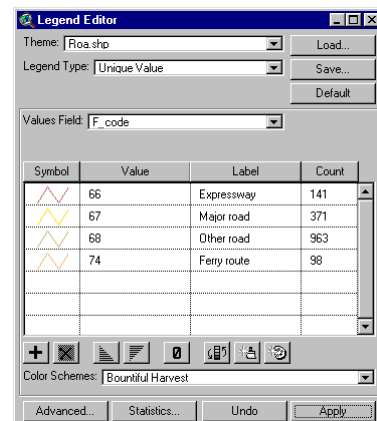
1. Click on the square next to the *roads* theme to make it active.
2. Use the **Zoom In** button to look closely at the high density areas.

- ? Do you see a relationship between the road network and the road density?**
- ? Do you think that all the roads of Canada are shown on this map?**

3. Double click **ecozone_trans** theme.
4. Open **Legend Type** list.
5. Select **Single Symbol**.
6. Double click **Symbol** box.
7. Open **Outline** list.
8. Select **None**.
9. Click **Apply**.
10. Close **Legend Editor** window.



11. Double click **Roa. (roads)** theme.
12. Open **Legend Type** list.
13. Select **Unique Value**.
14. Open **Values Field** list.
15. Select **F_code**.
16. Edit the **Label** list by changing the number code to the corresponding text listed below.
 - 66 Expressway
 - 67 Major road
 - 68 Other road
 - 74 Ferry route



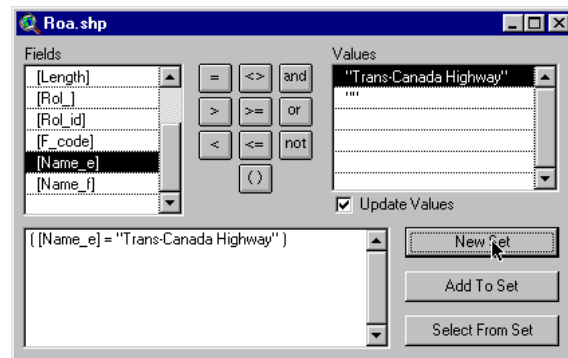
16. Click **Apply**.
17. Close **Legend Editor** window.
18. Use the **Zoom In** button to explore the road network.

- ? Explore the road network. Where are most of the expressways?
- ? Where are the road hubs?
- ? Which coast has more ferry routes?

ArcView - TRACING A NETWORK

Let's try and follow the Trans-Canada Highway across the country.

1. Double click **Roa. (roads)** theme.
2. Open **Legend Type** list.
3. Select **Single Symbol**.
4. Click **Apply**.
5. Close **Legend Editor** window.
6. Click **Query Builder**  button.
7. In the **Fields** list, double click **name_e**.
8. Click **=** button.
9. Double click **Trans-Canada Highway**.
10. Click **New Set**.
11. Close the **Query Builder** window.



- ? Is there more than one route for the Trans-Canada Highway?

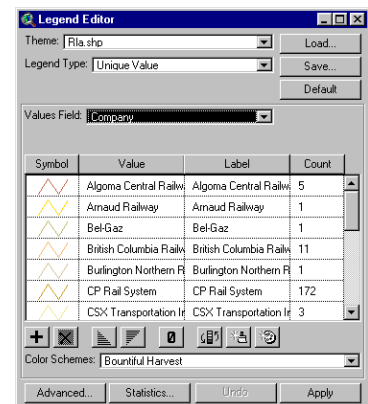
12. Click **Clear Selected Features**  button.
13. Uncheck **Roa. (roads)** theme to turn it off.

Now, let's look at the railroad network.

14. Click **Rla (railroads)** theme to activate it and check it to make it visible.
15. Use the **Zoom In** tool to explore the railroad network.

- ? Where do you find the largest number of railroad lines?
- ? Why?
- ? Where are the rail hubs?

16. Double click **Rla (railroads)** theme.
17. Open **Legend Type** list.
18. Select **Unique Value**.
19. Open **Value Field** list.
20. Select **company**.
21. Click **Apply**.



? How many railroad companies are there in Canada?

Click the  **Edit legend** button to get a count for the number of companies.

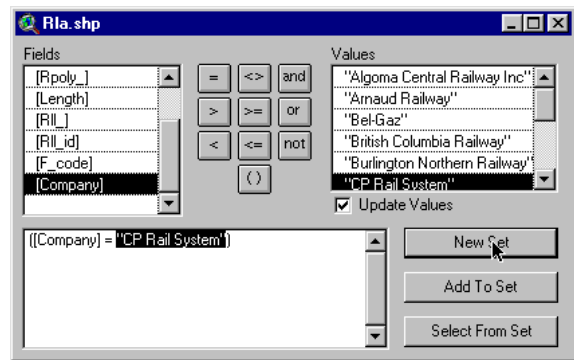
? Which are the two largest?

22. Close **Legend Editor** window.

ArcView - QUERYING THE DATABASE

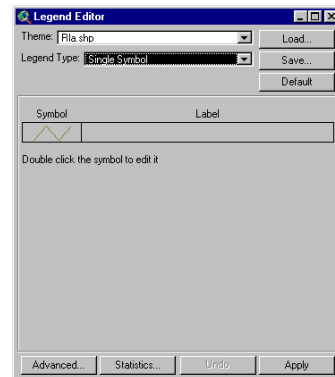
We have found that CP Rail System has a vast network of rail lines. Let's find all the rail lines that belong to Canadian Pacific.

1. Click the **Query Builder**  button.
2. Double click **company** in **Fields** list.
3. Click = button.
4. Double click **CP Rail System** in **Values** list.
5. Click **New Set**.
6. Close the **Query Builder** window.

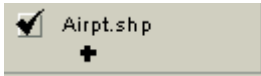


? Does CP Rail System operate in the Atlantic Provinces?

7. Double click **Rla (railroads)** theme.
8. Open **Legend Type** list.
9. Select **Single Symbol**.
10. Click **Apply**.
11. **Close Legend Editor** window.
12. Click **Clear Selected Features**  button.
13. Uncheck **Rla (railroads)** theme to turn it off.



We've looked at roads and railroads, now let's see where our airports are

located. 

1. Click on **Airpt (Airports)** theme to make it active.
2. Check **Airpt (Airports)** theme to make it visible.

? Can you describe the distribution of airports in Canada?
Explain.

3. Double click **Airpt (Airports)** theme.
4. Open **Legend Type** list.
5. Select **Unique Value**.
6. Select **Class**.
7. Click **Apply**.
8. Close **Legend Editor** window.



? Which provinces do not have international airports?

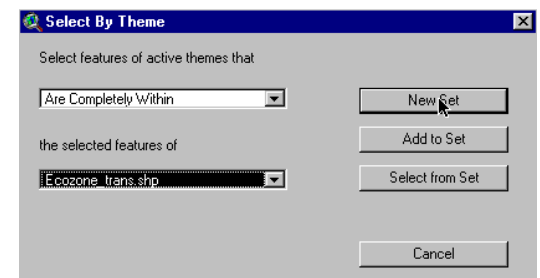
From our map, we can see that the majority of airports are located in the Mixedwood Plains ecozone. Let's find out how many.

1. Activate **ecozone_trans** theme by clicking on it.
2. Click **Open Theme Table**  button .
3. Click **Mixedwood Plains** in the table.
4. Close the table.



The **Mixedwood Plains** ecozone should be highlighted.

5. Click **Airpt (Airports)** theme to activate it.
6. Click MENU item **Theme**.
7. Click **Select By Theme**.
8. Open **Select features of active theme that list**.
9. Select **Are Completely Within**.
10. Open **the selected features of list**.
11. Select **ecozone_trans**.
12. Click **New Set**.



Airports within the **Mixedwood Plains** should be highlighted.

13. Click **Open Theme Table**  button.

- ? How many airports have been selected?
- ? What are their names?
- ? What class of airport are they?

ArcView - CREATING BUFFERS FOR COMPLEX QUERIES

Buffers are used to measure an area of a given distance surrounding a selected feature.

Examples would be the area of a circle surrounding a point 

or a worm shaped area surrounding a line feature.



Let's try a complex query. We want to find:

- how many population centres were within a 100 km radius of a selected airport
- how many people lived within that same radius in 1991.

1. Activate the **Airpt (Airports)** theme.

2. Click the **Select Features**  button.

3. Move the cursor over an airport you want to study and click. This will select that airport.

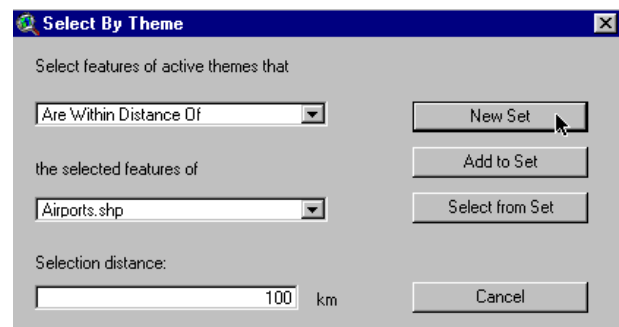
4. **Zoom In** if there are a number of airports close together.

5. Click **Open Theme Table**  and check that you have selected only one airport and record its name.

6. Close table.

ArcView - FINDING STATISTICS

- Click **pop_centers** theme to activate.
- Click MENU item **Theme**.
- Click **Select By Theme**.
- Open **Select features of active theme that list**.
- Select **Are Within Distance Of**.
- Open **the selected features of list**.
- Select **Airpt (Airports)**.
- Enter **100** in the **Selection distance**.
- Click **New Set**.

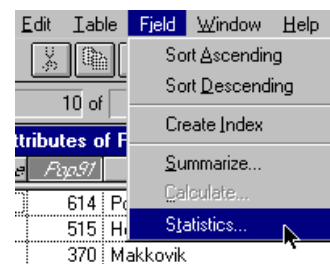


10. Click **Open Theme Table**  button.

11. Click the Field title **Pop91** to select the 1991 population field.

12. Click MENU item **Field**.

13. Click **Statistics**.



? How many population centres are within 100 km of your airport?

? How many people live within the 100 km radius of that airport?

Search the table to find the airport you selected and make a list of all the population centres that fall within 100 km.

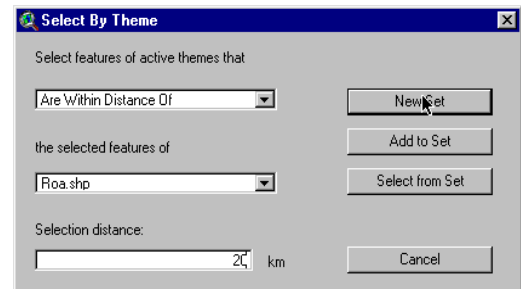
14. Close the table.

15. Check **pop_centers** theme.

? Looking at your map, can you see the distribution of population centres around your airport? Explain.

Let's use a buffer to determine how many people live within a 20 km area of the Trans-Canada Highway.

1. Activate the **Roa (roads)** theme. Click **Query Builder**  and select the **Trans-Canada Highway** as you did in the earlier part of this lesson.
2. Click **pop_centers** theme to activate.
3. Click MENU item **Theme** and **Select By Theme**.
4. In the **Selected features of active theme** that list, select **Are Within Distance Of**
5. In the **Selected feature of** list, select **Roa.shp**.
6. Enter **20** in **Selection distance**
7. Click **New Set**.
8. Close the window.
9. Click **Open Theme Table** button.
10. Click field title **Pop91** in the table.
11. Click MENU item **Field**.
12. Click **Statistics**.

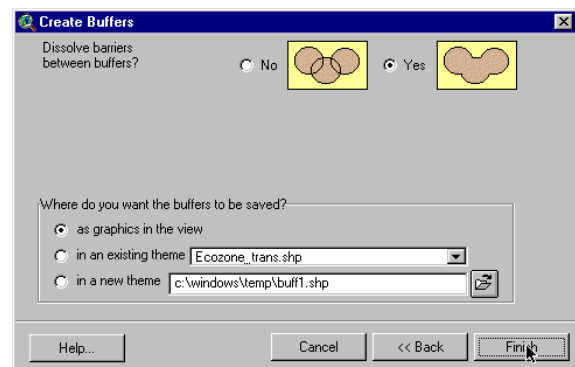


? How many population centres are located within 20 km of the Trans-Canada Highway?

? How many people are within this buffer?

ArcView - DRAW A BUFFER

1. Click **Roa (roads)** theme.
2. With the **Trans-Canada Highway** still selected, Click MENU item **Theme**.
3. Click **Create Buffers**.
4. Open **Theme** list.
5. Select **Roa (roads)**.
6. Click **Next**.
7. Click button **at a specific distance**.
8. Enter **20**.
9. Select **Distance units, Km**.
10. Click **Next**.
11. Next to **Dissolve Barriers** check **Yes**.
12. Save as **Graphics In View**.
13. Click **Finish**. This process may take a while. Be patient.



14. **Zoom In** on the Trans-Canada Highway to see a buffer.
15. Click on **Measure**  button.
16. Measure the distance from the highway to the edge of the buffer.

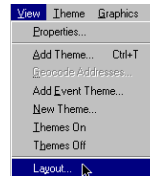
? Does zooming in give better readings?
 It should read 20 km.

ArcView - Map Layout for printing

Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

SET UP MAP

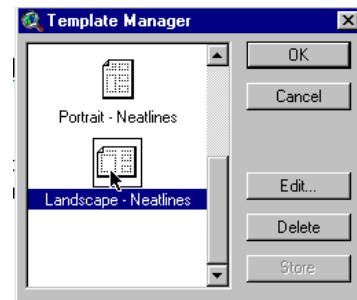
1. Click Menu **View**.
2. Click **Layout**.
3. Select the **Layout Template** you want to use.
 (One of the landscape templates is suggested).



4. Click **O.K.**
5. Maximize the Layout screen.

EDIT TITLE

1. Double click on the title **View1**.
2. Edit the title to reflect the purpose of the map, e.g., forestry, population.
3. Click **O.K.**
4. Resize the title using the corner handles and move it into a position that is visually balanced.
5. Click and drag to reposition the legend if necessary.



EDIT NORTH ARROW

1. Double Click on the *north arrow* and select a different presentation. Ensure that the arrow is pointing in the representative north direction of the map.
2. Click and drag to reposition or rotate (using rotation function) as needed.

EDIT SCALE BAR

1. Double click the *scale bar*.
2. Choose a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. Change the units to *kms*.
4. Find a combination of style, scale and divisions that is visually pleasing.
5. Click and drag the scale bar to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. Click on the **Text** tool.
2. Click at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., "Produced by (your name, date)".
4. Select desired formatting information such as **alignment**, **spacing** and **rotation**.
5. Click **O.K.** when it appears, as you want it.
6. Click on **Pointer** tool.
7. Select the new text.
8. Click once to change location or size, click twice to change content.

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

You only have to do this if you want to save your project on the computer or on a disk.

1. Click menu **Layout**.
2. Select **Properties**.
3. Change the "name". Click **O.K.**

ARRANGING YOUR LAYOUT

! Experiment with the layout buttons (use the tools page provided at the beginning of this lesson if you are unsure of their function).

- to **group** (tie pieces together so they move as one)
- to **ungroup** (break apart the ties so that individual pieces move)
- **bring to front** (change order that pieces were laid down, e.g, lines on top of polygons)
- **send to back** (change order that pieces were laid down, e.g, polygons behind lines)
- **zoom** (make image bigger or smaller)
- **neat line** (border around layout)

PRINT MAP LAYOUT

1. Click **Print** button.
2. Click **Setup**.
3. Select your printer properties as outlined by your teacher.
4. Click **O.K.** to print.

! **Note** that the map can also be saved to a printer file.